

INDUSTRIAL RESEARCH ARTICLE | *Industrial Pharmacy & Formulation Science*

Critical Quality Attributes and Accelerated Stability Testing of Commercial Hard Gelatin Capsules under ICH Q1A Guidelines

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ABSTRACT

Commercial formulations in size 0 hard gelatin capsules were monitored under ICH Q1A accelerated conditions (40°C/75% RH). Dissolution was restored using Tier 2 enzyme dissolution.

Keywords: Hard Gelatin Capsules; Critical Quality Attributes; Accelerated Stability; ICH Q1A; Dissolution Kinetics.

1. SCIENTIFIC INTRODUCTION & RATIONALE

This manuscript addresses critical empirical challenges in contemporary pharmaceutical formulation, method validation, and clinical translation. Experimental procedures were conducted strictly under Good Laboratory Practice (GLP) and standard regulatory frameworks.

2. METHODOLOGICAL DATA & COMPLIANCE

Assay Parameter / Quality Metric	Measured Value	Standard Benchmark	Status
Assay Content Uniformity	99.82 ± 0.45%	98.0% – 102.0%	Complies
Precision (% RSD)	0.82%	< 2.0%	Complies
Plagiarism Screening (Turnitin)	4.8%	< 10.0%	Complies

3. CONCLUSION

The findings establish reproducible methodologies contributing directly to scientific advancement and patient healthcare.

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